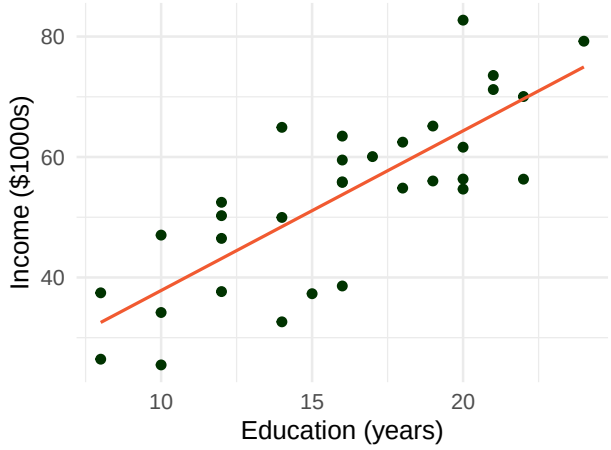
# Day 05: Evaluating SLR models

## Data for today

We return to the Riverview data (income vs. education) from last time




| term | estimate | std. error | t | p-value |
|---|---|---|---|---|
| Intercept | 11.321 | 6.123 | 1.85 | 0.074 |
| education | 2.651 | 0.370 | 7.17 | <0.001 |

## Two definitions of a ``good'' model

1. The model is appropriate, so our inferences (p-values, CIs) are _____
2. The x variable does a good job _____ the y variable

Theoretical model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ where $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$

Fitted model: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, so that $y_i = \hat{y}_i + \hat{\varepsilon}_i$

## Conditions for inference: LINE

For our p-values and confidence intervals to be trustworthy, the theoretical model above has to be a reasonable description of the data. Check:

- **L**inear: _____
- **I**ndependent and identically distributed errors: _____
- **N**ormally distributed errors: _____
- **E**qual variance of the errors: _____

> **Note**
>
> These conditions are about the *errors*, not about the raw x or y variables themselves.

## Residuals

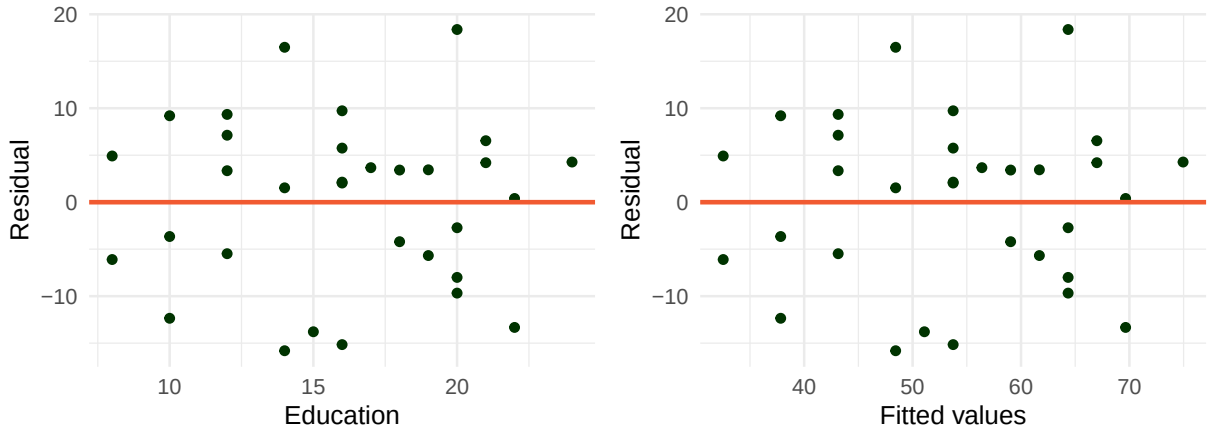
**Definition:** $e_i = \hat{\varepsilon}_i = y_i - \hat{y}_i$

**Properties:**

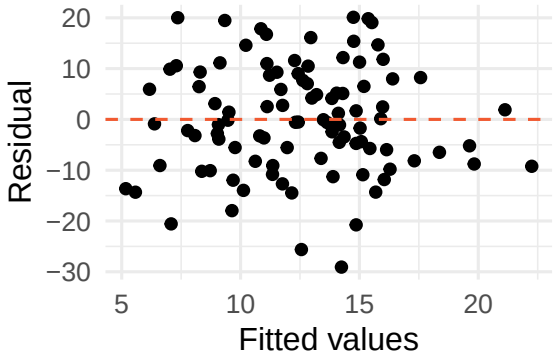
- sum to zero $\implies$ mean is ___
- **uncorrelated** with $x$ and $\hat{y}$
- (should be) normally distributed
- $SD(e_i) = \hat{\sigma}\sqrt{1 - \frac{1}{n} - \frac{(x_i - \overline{x})^2}{\sum(x_i - \overline{x})^2}}$

## Residual plots

For SLR, plotting residuals against $x$ or against the fitted values $\hat{y}$ shows the same shape (since $\hat{y}$ is a linear function of $x$) – either is a valid way to check linearity and equal variance.




**A _____ residual plot** shows no pattern: residuals bounce randomly around zero with roughly constant spread across the range of fitted values.



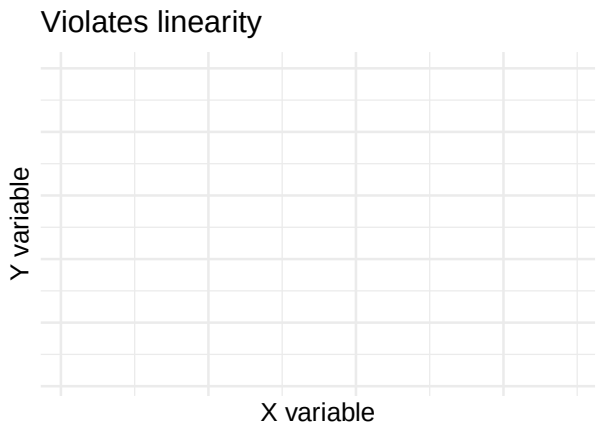
Group task

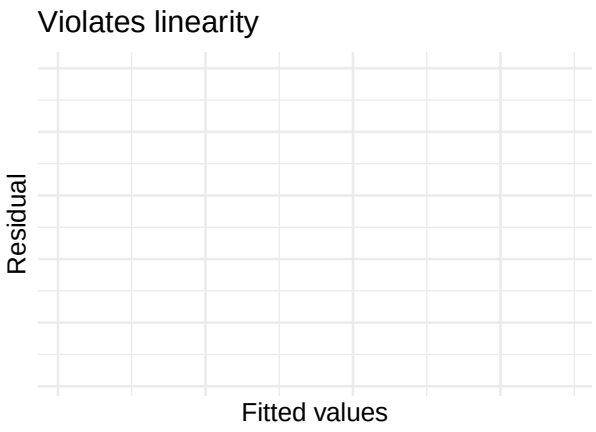
On the whiteboards, sketch a plot of $y$ vs. $x$ *and* the corresponding residual plot that would indicate a violation of

1. the linearity condition
2. the equal variance condition

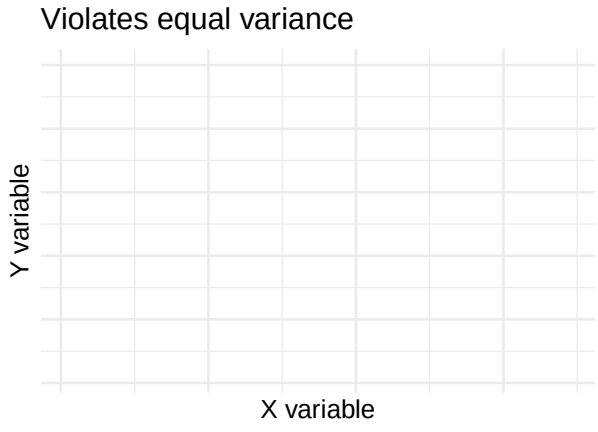
(wait to fill these in until after we discuss)

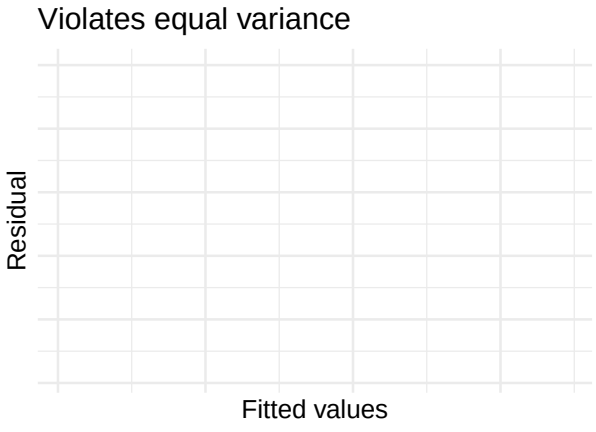
## Assessing Linearity



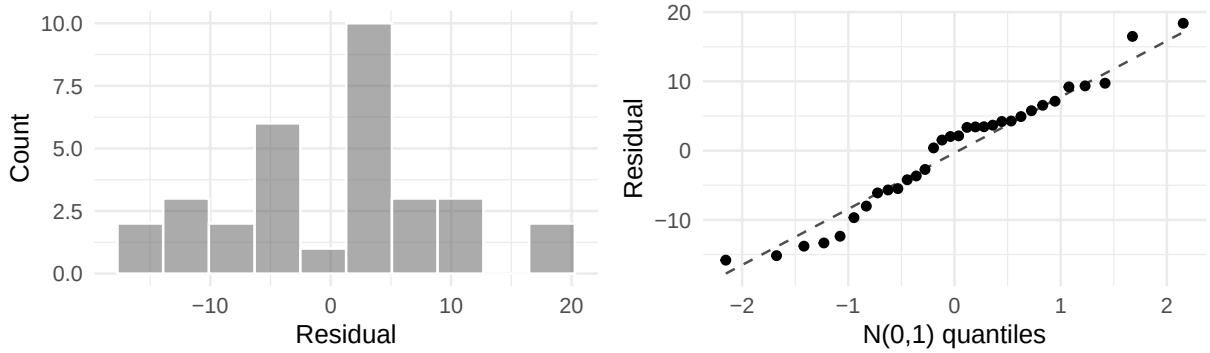

## Assessing Equal Variance



## Assessing Normality

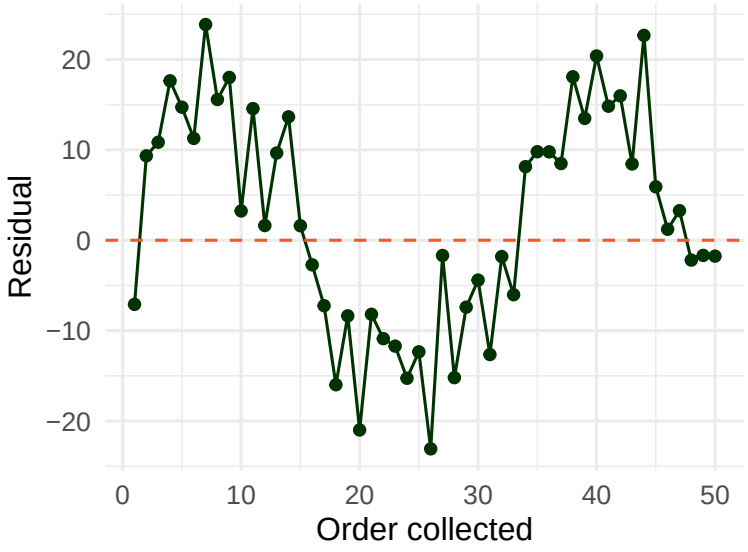
Use a histogram or normal Q-Q plot of the residuals. A "good" Q-Q plot hugs the diagonal reference line closely.




> **Note**
>
> Normality is **harder to judge at small sample sizes** – even true random draws from a normal distribution can look lumpy or have a wandering Q-Q plot when $n$ is small. Don't over-interpret minor wiggles with only 30-50 points.

## Assessing independence

Plot residuals against whatever variable might induce dependence (time, location, subject, batch, …). A clear pattern is a strong hint that the independence condition is violated. For example, if employees were hired in waves and residuals cycled up and down with the order they were added to the dataset, that would be a red flag:



## Making Residual Plots in R

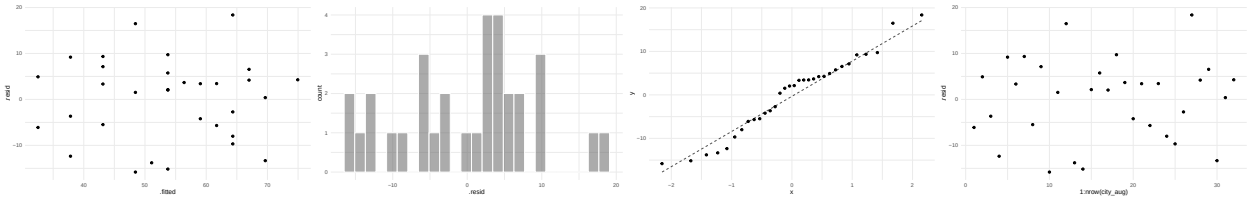
In order to make residual plots, we first need to find the residuals. While we can do this by hand for a couple of points, it would be extremely tedious to do it for *every single data point*! Instead, we'll use the `augment()` function from the {broom} R package to created an augmented dataset. This takes the original X and Y variables and adds the $\hat{y}$ values (`.fitted`) and residuals (`.resid`) (along with some other columns that we'll use later)

```
city_aug <- augment(city_lm)
city_aug
```

```
# A tibble: 32 x 8
   income education .fitted .resid   .hat .sigma .cooksd .std.resid
    <dbl>     <dbl>   <dbl>  <dbl>  <dbl>  <dbl>   <dbl>      <dbl>
 1   26.4         8    32.5  -6.10 0.140    9.05 0.0436      -0.733
 2   37.4         8    32.5   4.92 0.140    9.08 0.0283       0.590
 3   34.2        10    37.8  -3.65 0.0923   9.10 0.00927     -0.427
 4   25.5        10    37.8 -12.4  0.0923   8.81 0.106       -1.44
 5   47.0        10    37.8   9.20 0.0923   8.95 0.0588       1.08
 6   46.5        12    43.1   3.35 0.0584   9.11 0.00459      0.385
 7   52.5        12    43.1   9.34 0.0584   8.95 0.0356       1.07
 8   37.7        12    43.1  -5.48 0.0584   9.07 0.0123      -0.629
 9   50.3        12    43.1   7.13 0.0584   9.03 0.0207       0.818
10   32.6        14    48.4 -15.8  0.0380   8.63 0.0637      -1.80
# i 22 more rows
```

Once we have the augmented dataset, we can make scatterplots, etc. as usual with `gf_point`/`gf_histogram` etc.

```
gf_point(.resid ~ .fitted, data = city_aug)
gf_histogram(~.resid, data = city_aug, col = "white")
gf_qq(~ .resid, data = augment(city_lm)) |> gf_qqline()
gf_point(.resid ~ 1:nrow(city_aug), data = city_aug)
```



**Group task**

Try Part 1 of the R activity together

### What happens if the conditions aren't met?

- **Linearity:** if nonlinear, everything breaks!
- **Independence:** estimates are still unbiased, but the SEs are typically too **small** (we have less information than $n$ independent points suggests)
- **Normality:** estimates and SEs are still fine, but **prediction intervals** (which rely on individual points being normal around the mean) are wrong
- **Equal variance:** estimates are still unbiased, but SEs are wrong – and we don't know in which direction

### What do we do if assumptions are violated?

1. Change our assumptions (hard – needs more advanced methods)
2. Transform $y$, $x$, or both
3. Change the type of inference (e.g. bootstrap)

## Is x a ``good'' predictor of y?

### Partitioning variability




- SS _____: how far each point is from $\overline{y}$ (Total),
- SS _____: how far the model's prediction $\hat{y}_i$ is from $\overline{y}$
- SS _____: how far each point is from its own prediction $\hat{y}_i$

$$\underbrace{\sum (y_i - \overline{y})^2}_{SST} = \underbrace{\sum (\hat{y}_i - \overline{y})^2}_{SSR} + \underbrace{\sum e_i^2}_{SSE}$$

total variability $=$ variability of $\hat{Y}$'s $+$ variability of residuals (_____)

$$R^2 = \frac{SSR}{SST} = \frac{\text{variability of } \hat{Y}}{\text{variability of } Y}$$

**$R^2$**

$R^2$ gives the proportion of total variability in $Y$ explained by the SLR model.

For the Riverview data, $R^2 = 0.632$ – about 63.2% of the variability in income is explained by _____.

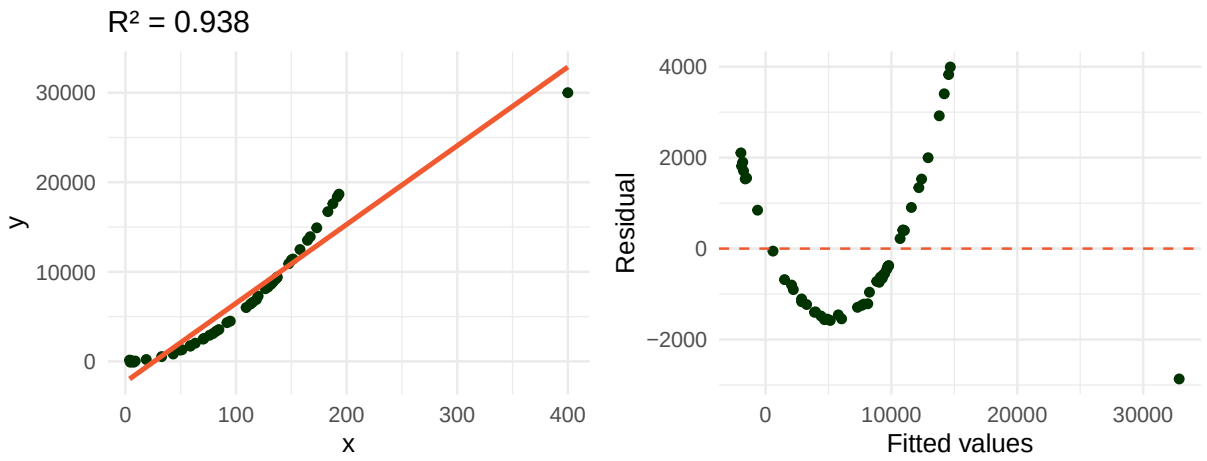
## Properties of $R^2$

- Between ___ and ___
- Big when $\sigma$ is small relative to the overall variability of $y$ (i.e. a big proportion of variability explained by the model)
- For SLR, $R^2 =$ ___ (where $r$ is the correlation coefficient between $x$ and $y$)

**Quick notation reference**

| Symbol | Meaning |
|---|---|
| $r$ | sample correlation coefficient |
| $\rho$ | population correlation coefficient |
| $R^2$ | coefficient of determination |

## $R^2$ dangers:




**Your turn**

Would you trust this model just based on its $R^2$? What does the residual plot tell you that $R^2$ alone can't?

## Finding $R^2$ in R

We get $R^2$ as part of the summary() output:

```
summary(city_lm)
```

```
Call:
lm(formula = income ~ education, data = city)

Residuals:
    Min      1Q  Median      3Q     Max
-15.809  -5.783   2.088   5.127  18.379

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.3214     6.1232   1.849   0.0743 .
education     2.6513     0.3696   7.173 5.56e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.978 on 30 degrees of freedom
Multiple R-squared:  0.6317,    Adjusted R-squared:  0.6194
F-statistic: 51.45 on 1 and 30 DF,  p-value: 5.562e-08
```

and you can access it directly with $:

```
summary(city_lm)$r.squared
```

```
[1] 0.6316828
```